

Dual CNN for photovoltaic electroluminescence images microcrack detection

Khoulood Samrouth^{a, ID, *}, Souha Nazir^b, Nader Bakir^c, Nadine Khodor^d

^a Arab Open University, Beirut, Lebanon

^b Ivanae Medical, Brest, France

^c Beirut Arab University, Tripoli, Lebanon

^d ESIEA, Paris, France

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ABSTRACT

Accurate detection of microcracks in photovoltaic (PV) cells is crucial for ensuring the efficiency and longevity of solar panels. This study proposes a dual convolutional neural network (Dual-CNN) architecture to enhance microcrack detection in electroluminescence (EL) PV images. By integrating shallow feature extraction with deep semantic analysis, the proposed model effectively captures both fine-grained local textures and high-level structural patterns, addressing the limitations of conventional single-stream CNN models that primarily focus on coarse-grained features. Experimental evaluations on an EL image dataset demonstrate that the Dual-CNN approach significantly improves defect localization and classification with an accuracy of 85.33%, a recall of 71.71% and an F1 score of 73.9%, paving the way for more robust and automated PV inspection systems in the solar energy sector.

1. Introduction

Given the worldwide ongoing economic crisis and frequent power outages, there is a growing focus on environmental sustainability and the adoption of renewable solar energy systems [1] aiming to combat climate change, reduce carbon emissions, and transition away from fossil fuels. Photovoltaic (PV) modules are particularly becoming widespread in residential, commercial, and industrial settings to harness clean energy [2,3].

A solar cell, also known as a photovoltaic (PV) cell and shown in Fig. 1a, is a device that converts light energy, typically from the sun, into electrical energy through the photovoltaic effect. It serves as the fundamental component in solar energy systems, enabling the generation of clean, renewable power [4].

One significant challenge in maintaining PV module efficiency is the detection of microcracks. These microcracks, though subtle in appearance, can progressively worsen over time, eventually compromising the module's efficiency. While identifying the exact location of these defects within a solar cell may be less critical for the overall quality assessment of a PV module, detecting them early remains essential to prevent long-term performance degradation.

To identify such hidden issues that can affect the performance and efficiency of solar panels, Electroluminescence (EL) imaging technique is widely used [5]. As illustrated in Figs. 1b and 1c, EL PV image is a visual representation of a PV module's internal structure, captured

by detecting the light emitted from the cells when a forward bias is applied. These images highlight defects such as microcracks (1.c), shunts, and other irregularities within the PV cells that are invisible in standard visual inspections. The contrast in the image indicates areas of potential concern, with darker regions typically signaling defects.

On one side, Deep Learning (DL) plays an important role in crack detection specifically in Concrete Building Structures [6–8] and roads [9]. For instance, authors in [6–8], enhanced crack identification in concrete buildings by processing more input pixels and combining convolution channel attention with window-based self-attention mechanisms. This dual stream approach has been also used in [9] to improve the crack localization in the road using vision transformer (ViT) and convolutional neural networks (CNNs)-based encoder and decoder.

On the other hand, DL plays a crucial role in enhancing the efficiency and reliability of renewable green energy systems [10,11], particularly in the context of image processing tools. By integrating deep learning techniques like Convolutional Neural Networks (CNNs), researchers are able to make precise forecasts of future power usage and renewable energy installations, ultimately improving power control assistance and the integration of renewable energy sources [12]. These advancements are essential in addressing the challenges posed

* Corresponding author.

E-mail addresses: ksamrouth@aou.edu.lb (K. Samrouth), souha.nazir@ivanae-medical.com (S. Nazir), nbakir@bau.edu.lb (N. Bakir), nadine.khodor@esiea.fr (N. Khodor).

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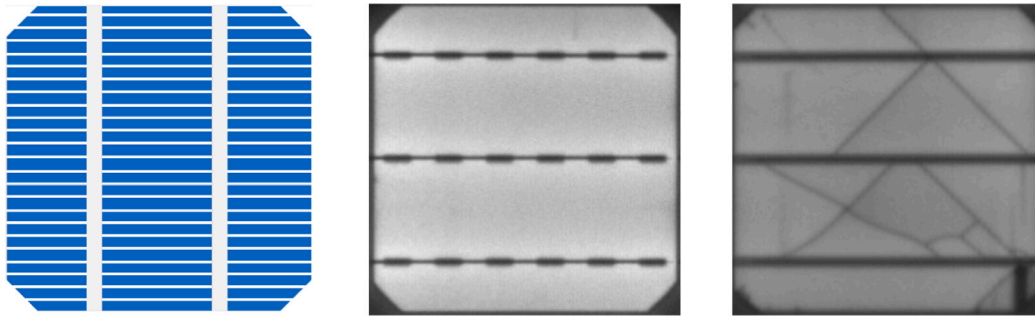


Fig. 1. (a) PV cell. (b) EL image of normal PV cell. (c) EL image of cracked PV cell.

by the volatility, intermittence, and uncertainty of renewable energy resources, ultimately contributing to the stability and reliability of large-scale renewable energy integration into power generation. Furthermore, the utilization of machine learning and deep learning approaches in renewable energy prediction models has proven to be effective in discovering nonlinear characteristics and high-level invariant structures in data, thus enhancing the reliability of renewable energy predictions for power-system scheduling, operations, and management.

However, existing DL techniques for detecting defects in EL PV images face several limitations. Traditional DL methods often struggle to accurately detect subtle microcracks and other minor defects due to their reliance on either shallow or deep feature extraction alone, which can result in missed or inaccurate detections [13].

Beyond CNN-based approaches, Vision Transformers (ViTs) have recently demonstrated remarkable performance in computer vision tasks by utilizing self-attention mechanisms to model long-range dependencies in images. Unlike traditional CNNs, which rely on local receptive fields, ViTs can process entire EL images holistically, allowing them to identify microcracks that may span across multiple regions. By integrating ViTs with CNN-based architectures, hybrid models can further enhance defect detection by combining the spatial feature extraction capabilities of CNNs with the global attention mechanisms of transformers. Recent studies have explored transformer-based methods for defect detection in industrial applications, showing their potential to improve accuracy and robustness compared to conventional DL approaches [14,15].

Our proposed method using a dual convolutional neural network (Dual CNN) [16] addresses these limitations by combining the strengths of both shallow and deep feature extraction. The dual CNN approach consists of 2 networks. The first network captures fine-grained, local features, while the second network extracts deep semantic information. This integration enhances the model's ability to detect subtle defects and improve overall accuracy, making it more robust to varying conditions and better suited for precise quality assessment in EL PV images.

In the following, we highlight the key contributions of our method:

- **Dual-Stream Feature Extraction:** The proposed Dual-CNN architecture effectively integrates shallow and deep feature extraction, capturing both fine-grained local textures and high-level semantic patterns for improved microcrack detection.
- **Enhanced Microcrack Detection:** By combining two parallel networks, the model significantly improves the detection of subtle microcracks, which are often missed by traditional CNNs.
- **Robustness to Varying Conditions:** The architecture enhances defect detection performance across diverse EL PV images, ensuring reliability under different noise levels and imaging conditions. By combining local and global features, the model can effectively suppress irrelevant noise while retaining crucial defect-related features, improving detection accuracy.
- **Improved Classification and Localization:** The integration of local and global features enables more precise defect classification and better localization of microcracks.

- **Scalability for Automated Inspection:** The method is well-suited for real-world deployment, contributing to the development of automated PV inspection systems for large-scale solar panel assessment.

The remainder of this paper is organized as follows. In Section 2, we analyze the existing deep learning-based image processing tools for PV crack detection. In Section 3, we detail our proposed Dual CNN based detection PV crack method. Finally, in Section 7, we conclude this paper.

2. Related works

Traditional pattern recognition methods were widely utilized in the literature to detect cracks and other defects in PV images. Techniques such as Fourier image reconstruction [17] have been applied to analyze frequency components in images, aiming to identify periodic patterns that correspond to defects. Independent component analysis (ICA) [18] has been employed to separate mixed signals in PV images, isolating independent sources of variation that may indicate cracks or other anomalies. Anisotropic diffusion filtering [19] has also been used to enhance edge detection by smoothing image regions while preserving edges, which helps in highlighting the boundaries of defects. However, these traditional techniques often face challenges in accurately identifying subtle and complex defects due to their reliance on handcrafted features and sensitivity to noise and image artifacts. Moreover, these methods typically require significant pre-processing and are often limited by their inability to generalize across different types of defects and imaging conditions.

With the advent of deep learning, more sophisticated approaches have been developed to address these limitations. Deep learning techniques, particularly convolutional neural networks (CNNs), have revolutionized the field by significantly enhancing the detection and categorization of defects in PV panels. These methods have been successfully applied to various tasks, including defect detection, dust detection, maintenance optimization, and sun tracking, all of which contribute to improving the overall efficiency of solar panels.

For instance, CNN-based models such as VGG16, VGG19, Inception, ResNet50, Xception, EfficientNet, U-Net, and customized CNN architectures have been utilized to analyze both infrared aerial images and satellite imagery [20–26]. These models are capable of identifying and categorizing defects such as cell damage, diode malfunctions, and hotspots with high accuracy. The use of transfer learning with pre-trained networks like VGG19 and ResNet50 has also been explored, enabling models to leverage learned features from large image datasets, thus improving their ability to detect subtle defects that might be missed by traditional methods.

Additionally, ensemble learning techniques, which combine multiple models to improve predictive performance, have been used to enhance the robustness and accuracy of defect detection in PV images. Researchers have developed sophisticated methods that integrate CNNs with other techniques like U-Net, which is particularly effective in

segmenting defects in EL images [27–30]. These deep learning models not only enable the rapid and accurate identification of various defects but also support the development of automated maintenance strategies. By detecting issues early, such as cell cracks or hotspots, these methods contribute to reducing operational costs and maximizing energy production, ultimately leading to greater efficiency and cost savings in solar energy systems.

Overall, deep learning approaches offer advanced capabilities to identify and analyze defects more accurately and efficiently than traditional methods.

However, existing DL techniques for detecting defects in EL PV images face notable challenges. Traditional DL methods often fail to accurately detect subtle microcracks and other minor defects because they rely solely on either shallow or deep feature extraction, leading to missed or incorrect detections.

3. Proposed method

In contrast with the existing DL-based PV cells assessment techniques, our proposed method is based on a dual convolutional neural network (Dual CNN) that combines the advantages of both shallow and deep feature extraction. In particular, the Dual CNN comprises two parallel convolutional neural networks (CNNs): the first captures fine-grained, local features, while the second extracts deep semantic information. This combination enhances the model's ability to detect subtle defects, improving overall accuracy and robustness to varying conditions, making it ideal for precise quality assessment in EL PV images.

We employed a dual convolutional neural network (Dual CNN) model architecture [31], as illustrated in Fig. 2 and briefly described in Algorithm 1. It typically consists of two parallel CNN streams that process the input data simultaneously but focus on different aspects of feature extraction. The outline of our model is as follows:

- **EL Image Normalization:** The input grayscale EL images, each with a resolution of 300×300 pixels, are normalized to a pixel value range of 0 to 1, according to Eq. (1), where $\hat{x}$ represents the normalized pixel value, x is the original pixel value, 255 is the maximum pixel value over 8 bits. This normalization enhances both the speed and robustness of the processing.

$$\hat{x} = \frac{x}{255} \quad (1)$$

Algorithm 1 Dual CNN for PV Defect Detection

- 1: **Input:** PV image I
- 2: **Output:** Predicted defect class C
- 3: **Preprocessing:**
- 4: Resize I to 300×300
- 5: Normalize pixel values
- 6: Apply data augmentation (rotation, flipping)
- 7: **Feature Extraction Using Dual CNNs:**
- 8: CNN Branch 1: Extract low-level spatial features (edges, textures)
- 9: CNN Branch 2: Extract high-level semantic features (defect patterns)
- 10: **Feature Fusion:**
- 11: Concatenate feature maps from both CNN branches
- 12: **Classification Head:**
- 13: Pass combined features through fully connected layers
- 14: Apply softmax activation to obtain class probabilities
- 15: **Prediction:**
- 16: Assign class C with the highest probability as the defect classification

- Dual CNN Streams:

- **Shallow CNN Stream:** The shallow CNN stream is designed to capture fine-grained, local features by utilizing a minimal number of convolutional layers. Specifically, it consists of two convolutional layers with small kernel sizes (3×3), each followed by a max pooling layer. This lightweight architecture is beneficial because early convolutional layers in deep networks tend to preserve local texture and edge information, which are critical for distinguishing subtle defect patterns in PV images. By limiting the depth of the network, the model emphasizes localized spatial information without abstracting it away through deeper hierarchical representations. The use of max pooling after each convolutional layer helps reduce the spatial dimensions while retaining the most salient features, improving efficiency and reducing overfitting. The final feature map is flattened and passed to a fully connected layer with 128 neurons, activated by a ReLU function, which enables non-linear decision boundaries. Table 1 summarizes the architecture of the shallow CNN stream.

- **Deep CNN Stream:** This stream captures more abstract, deep semantic features through a deep CNN-based model. In particular, we tested 3 deep CNN-based models independently.

- * **Custom deep convolutional network:** Our customized deep CNN stream is composed of 3 convolutional layers with larger kernels (7×7 , 5×5 , and 3×3 , respectively) and an increasing number of filters (64, 128, and 256, respectively), each followed by max pooling layers as illustrated in Fig. 3. It aims to capture higher-level patterns and context from the image. The last feature map is flattened through a flatten layer and fed to a fully connected network of 512 neurons with a ReLU activation function. Table 2 summarizes the architecture of our custom deep CNN stream.
- * **VGG19:** a deep convolutional neural network known for its simple yet effective architecture, consisting of 19 layers that emphasize small convolutional filters to capture fine details in images. It has been widely used in computer vision tasks due to its balanced depth and computational efficiency [32].
- * **AlexNet:** deep learning model with eight layers [33].

It is worth noting that we experimented with ResNet and EfficientNetB0; however, their performance did not meet expectations. Therefore, we focused solely on the three aforementioned deep CNN models for further analysis.

- **Feature Fusion Layer:** The outputs from both CNN streams are simply concatenated through the feature fusion layer.
- **Fully Connected Layers:** The combined features are then passed through 1 fully connected layer of 256 neurons with a ReLU activation function to integrate and interpret the combined information. Table 3 summarizes the architecture of the fully connected layers in our model.
- **Output Layer:** The final output layer, presented in Table 3, produces the binary classification (defect or no defect) with 1 neuron and a Sigmoid activation function.
- **Loss Function and Optimization:** The model is trained using the binary cross entropy loss function and an adaptive learning rate optimization algorithm (Adam [34]) to minimize the error between the predicted and actual values. Adam combines the advantages of two other popular optimizers: AdaGrad and RMSProp. It uses first-order gradients with adaptive estimates of lower-order moments to adjust the learning rate for each parameter dynamically according to Eq. (2), where: η is the learning rate, $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected first and second moment estimates, and ϵ is a small constant to prevent division by zero.

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} \hat{m}_t \quad (2)$$

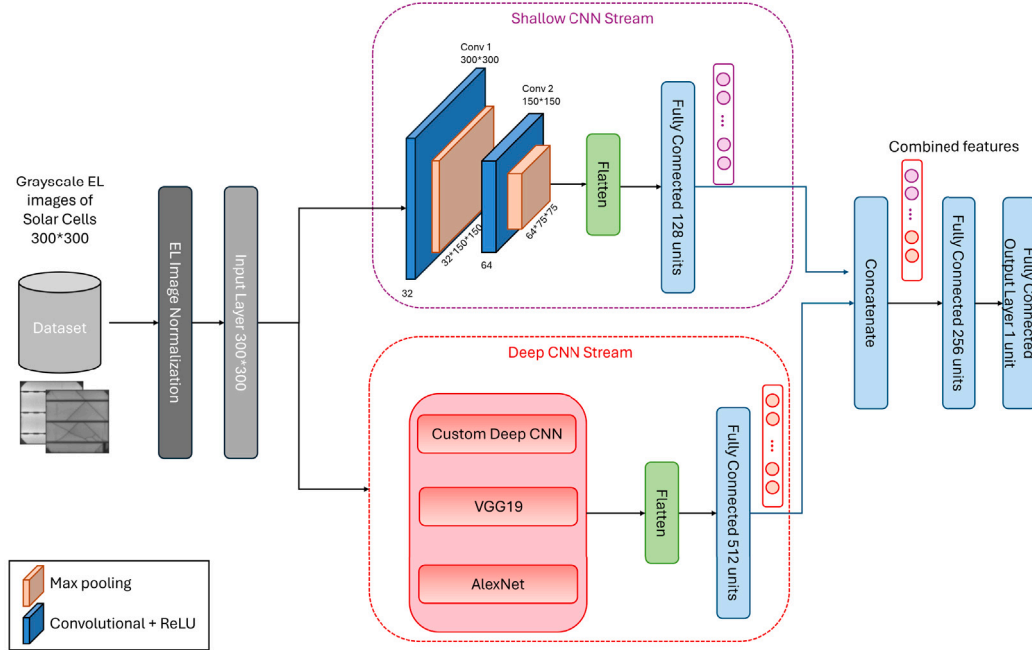


Fig. 2. Proposed Framework based on Dual CNN architecture.

Table 1
Shallow CNN Stream Architecture.

Layer	Configuration	Output Size
Input	Image (300 × 300)	300 × 300
Convolutional Layer	32 filters, 3 × 3 kernel, ReLU	300 × 300 × 32
Max Pooling Layer	2 × 2 pool size, stride 2	150 × 150 × 32
Convolutional Layer	64 filters, 3 × 3 kernel, ReLU	150 × 150 × 64
Max Pooling Layer	2 × 2 pool size, stride 2	75 × 75 × 64
Flatten	Flatten (75 × 75 × 64)	360,000 units
Fully Connected Layer	128 units, ReLU	128 units

Table 2
Our Custom Deep CNN Stream Architecture.

Layer	Configuration	Output Size
Input	Image (300 × 300)	300 × 300
Convolutional Layer	32 filters, 7 × 7 kernel, ReLU	300 × 300 × 32
Max Pooling Layer	2 × 2 pool size, stride 2	150 × 150 × 32
Convolutional Layer	64 filters, 7 × 7 kernel, ReLU	150 × 150 × 64
Max Pooling Layer	2 × 2 pool size, stride 2	75 × 147 × 64
Convolutional Layer	128 filters, 5 × 5 kernel, ReLU	143 × 143 × 128
Max Pooling Layer	2 × 2 pool size, stride 2	71 × 71 × 128
Convolutional Layer	256 filters, 3 × 3 kernel, ReLU	69 × 69 × 256
Max Pooling Layer	2 × 2 pool size, stride 2	34 × 34 × 256
Flatten	Flatten (34 × 34 × 256)	295,936 units
Fully Connected Layer	512 units, ReLU	512 units

Table 3
Feature Fusion and Final Layers Architecture.

Layer	Configuration	Output Size
Feature Fusion	Concatenation of shallow and deep CNN features	512+128=640
Fully Connected Layer	256 units, ReLU	256 units
Output Layer	1 unit, Sigmoid	1 unit

4. Proposed model mathematical formulation

The proposed Dual-CNN model consists of two parallel convolutional neural networks (CNNs):

- **Shallow CNN (F_s)** for extracting fine-grained, local features.
- **Deep CNN (F_d)** for extracting high-level semantic features.

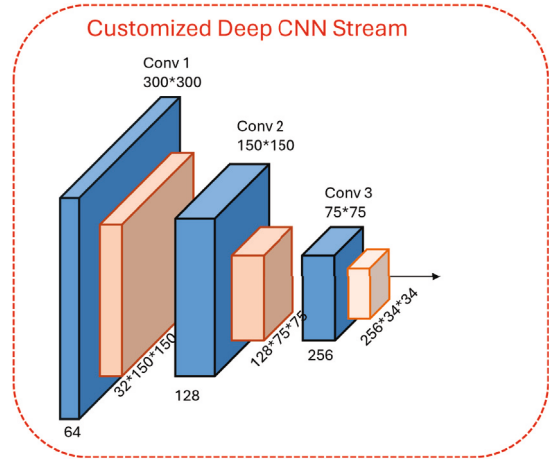


Fig. 3. Custom Deep CNN architecture.

- Feature fusion strategy to enhance defect detection.

4.1. Input representation

Let $X \in \mathbb{R}^{H \times W \times C}$ represent an input EL PV image, where:

- H = Image height
- W = Image width
- C = Number of channels (grayscale: $C = 1$)

The image is normalized using:

$$\tilde{X} = \frac{X - \mu_X}{\sigma_X} \quad (3)$$

where:

- μ_X = Mean pixel intensity of X
- σ_X = Standard deviation of pixel intensities

4.2. Shallow feature extraction (F_s)

The shallow CNN extracts fine-grained features:

$$F_s = F_s(\tilde{X}) = \sigma(W_s * \tilde{X} + b_s) \quad (4)$$

where:

- $F_s \in \mathbb{R}^{H' \times W' \times D_s}$ is the shallow feature map.
- W_s are the convolutional kernel weights.
- b_s is the bias term.
- $*$ denotes the convolution operation.
- $\sigma(\cdot)$ is the activation function (ReLU).
- D_s is the number of filters in the last shallow CNN layer.

4.3. Deep feature extraction (F_d)

The deep CNN extracts high-level semantic features:

$$F_d = F_d(\tilde{X}) = \sigma(W_d * \tilde{X} + b_d) \quad (5)$$

where:

- $F_d \in \mathbb{R}^{H'' \times W'' \times D_d}$ is the deep feature map.
- W_d are the convolutional kernel weights for deep CNN.
- b_d is the bias term.
- D_d is the number of filters in the last deep CNN layer.

4.4. Feature fusion

The shallow and deep feature maps are fused using a weighted sum:

$$F_{\text{fusion}} = \text{Concat}(F_s, F_d) \quad (6)$$

where:

- $F_{\text{fusion}} \in \mathbb{R}^{H'' \times W'' \times (D_s + D_d)}$ is the fused feature representation.
- α, β are learnable weighting factors for balancing shallow and deep features.

4.5. Classification layer

The fused features are passed through a fully connected (FC) layer:

$$y = \text{softmax}(W_{\text{fc}}(\text{Flatten}(F_{\text{fusion}})) + b_{\text{fc}}) \quad (7)$$

where:

- $y \in \mathbb{R}^K$ represents the probability distribution over K classes (microcrack vs. no microcrack).
- W_{fc} are the weights of the fully connected layer.
- b_{fc} is the bias term.

The softmax function is defined as:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^2 e^{z_j}} \quad (8)$$

4.6. Loss function (Cross-Entropy Loss)

The model is optimized using the categorical cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^2 y_i \log(\hat{y}_i) \quad (9)$$

where:

- y_i is the ground truth label for class i .
- $\hat{y}_i$ is the predicted probability for class i .

Table 4

Summary of the key variables used in the Dual-CNN model.

Variable	Definition
X	Input EL PV image ($\mathbb{R}^{H \times W \times C}$)
$\tilde{X}$	Preprocessed image
F_s	Shallow feature map
F_d	Deep feature map
F_{fusion}	Combined feature representation
α, β	Learnable fusion weights
y	Predicted class probabilities
K	Number of output classes (binary: defect/no defect)
W_s, W_d	Convolutional weights for shallow/deep CNN
b_s, b_d	Bias terms
$W_{\text{fc}}, b_{\text{fc}}$	Fully connected layer weights and bias
$\mathcal{L}$	Loss function (cross-entropy)
η	Learning rate
B	Batch size

Table 5

The distribution of the total number of solar cell images in the dataset. The numbers of solar cell images are given for the 60% training, 20% validation, 20% testing.

Number of images	Training	Testing	Validation
	1574	525	525

4.7. Training process

The model is trained using:

- **Optimization Algorithm:** Adam optimizer
- **Learning Rate:** $\eta = 10^{-4}$
- **Batch Size:** $B = 32$
- **Regularization:** L_2 weight decay applied to convolutional and fully connected layers

Table 4 summarizes the key variables of our proposed method.

5. Experiments and results

To evaluate our approach, we implemented several CNN-based models, including our custom deep CNN model, VGG19, and AlexNet, in 2 setups: single-stream setup and in Dual stream setup where each model (custom deep DNN, VGG19, or AlexNet) is coupled with the shallow CNN stream described in Section 3.

5.1. Development environment

We conducted our experiments using the TensorFlow framework on Windows 11 Pro operating system, Python 3.12.4, and an Nvidia GeForce RTX 4060 Laptop GPU with 16 GB RAM. We made our code publicly available on <https://github.com/ksamrout/Dual-CNN-for-Photovoltaic-Electroluminescence-images-MicroCrack-Detection>

5.2. Datasets

We used an open source dataset that contains 2624 samples of 300*300 8-bit grayscale EL images of functional and defective solar cells with varying degree of degradations extracted from 44 different solar modules [28,35,36]. We applied the 80%, 10%, 10% splitting for training, validation and testing, respectively. Therefore, we got 1574 EL images reserved for training, 525 EL images reserved for validation and 525 EL image for testing, as shown in Table 5.

For further testing, we used a second dataset that consists of 3550 samples of 8-bit grayscale EL images of good and cracked/corroded solar cells [37].

We augment our dataset using morphological rotation and several noise augmentation techniques: Gaussian noise, salt-and-pepper noise, and contrast variations, that imitate real-world imaging conditions, as follows:

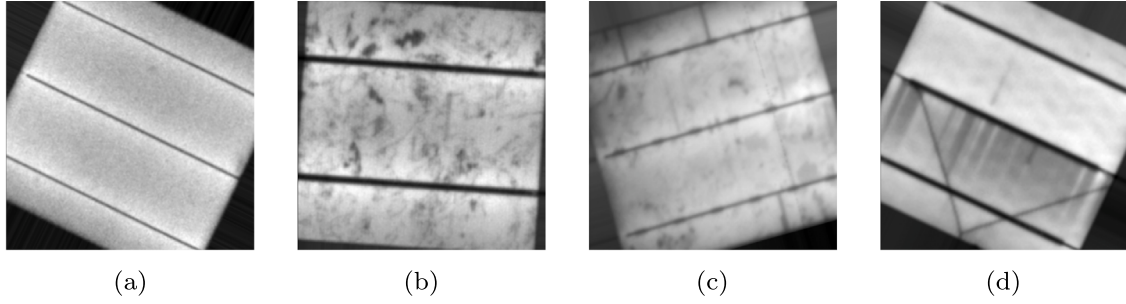


Fig. 4. Examples of augmented data samples including a combination of rotation, salt & pepper noise and contrast variation.

- Rotation: Applied with a maximum angle of rotation of 30 degrees.
- Gaussian noise: Added with a mean of 0 and varying standard deviations ($\sigma = 0.01, 0.05, 0.1$) to simulate sensor noise.
- Salt-and-pepper noise: Introduced with noise densities ranging from 1% to 5%, mimicking pixel corruption due to transmission errors or faulty sensors.
- Contrast variation: Achieved using random contrast stretching and histogram equalization to account for lighting inconsistencies across different imaging scenarios.

We apply these augmentations stochastically during training to increase model robustness and generalization. In addition, we implemented all these augmentation routines using Tensorflow and they are reproducible via fixed random seeds. Fig. 4 shows several examples of augmented data samples including a combination of rotation, salt & pepper noise and contrast variation.

5.3. Training

Although TensorFlow abstracts away much of the low-level optimization details, our training process is grounded in standard optimization theory. The objective is to minimize the binary cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (10)$$

where y_i is the ground-truth label and $\hat{y}_i$ is the predicted probability for sample i . The loss is minimized using the ADAM optimizer [34], which adaptively updates model parameters using estimates of first and second moments of the gradients. While the exact update equations are handled internally by TensorFlow, we configured the optimizer with a learning rate of 10^{-4} , and used default hyperparameters ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$).

We train our model through 15 epochs with a batch size of 28. To prevent stagnation during training, we employed the ReduceLROnPlateau scheduler. This monitors the validation loss and reduces the learning rate by a factor of 0.2 if no improvement is observed over 5 consecutive epochs, with a minimum learning rate threshold of 10^{-6} . This strategy encourages better convergence in the non-convex loss landscape typical of deep neural networks.

Figs. 5a, 6a and 7a show the evolution of the training and validation loss and accuracy, respectively, through 15 epochs for the custom deep CNN, VGG19 (with frozen backbone layers parameters) and AlexNet in a single stream setup.

Figs. 5b, 6b and 7 show the evolution of the training and validation loss and accuracy, respectively, through 15 epochs for the custom deep CNN, VGG19 (with frozen backbone layers parameters), and AlexNet in a dual stream setup.

These training and validation curves show stable convergence and effective learning. In addition, they show that with more training epochs, the models still have room for better performance. Moreover, the small gap between training and validation loss indicates good generalization.

5.4. Classification results

To evaluate the performance of our model, we conducted an ablation study by comparing the proposed dual CNN architecture with a single-stream deep architecture on different CNN-based models (Custom CNN, VGG19, and AlexNet). This comparison allows us to isolate the impact of the dual-stream architecture on the overall performance. We performed a comprehensive analysis using both the confusion matrix and Receiver Operating Characteristic (ROC) curves to compare the classification accuracy, sensitivity, and specificity of both architectures, highlighting the improvements gained by the dual CNN approach.

Fig. 8a shows the confusion matrix for the single-stream custom deep CNN model with 319 True Positives (TP), 106 True Negatives (TN), 50 False Negatives (FN) and 50 False Positives (FP). Fig. 8b shows the confusion matrix of our proposed dual custom CNN classification model with 339 TP, 109 TN, 43 FN and 34 FP.

Fig. 8c shows the confusion matrix for the single-stream VGG19 model with 329 TP, 99 TN, 57 FN and 40 FP. Fig. 8d shows the confusion matrix of VGG19 classification model in the dual setup with 339 TP, 109 TN, 47 FN and 30 FP.

Fig. 8e shows the confusion matrix for the single-stream AlexNet model with 343 TP, 81 TN, 75 FN and 26 FP. Fig. 8f shows the confusion matrix of the AlexNet classification model in dual stream setup with 353 TP, 91 TN, 65 FN and 16 FP.

These confusion matrices reveal a clear performance improvement in the dual-stream setup across all tested models compared to their single-stream counterparts.

- Custom Deep CNN: In the single-stream setup (Fig. 8a), this model achieves 319 TP and 106 TN, with 50 FN and 50 FP. In the dual-stream setup (Fig. 8b), TPs and TNs increase to 339 and 109, respectively, with reduced FNs (43) and FPs (34). This indicates an enhancement in both sensitivity (fewer FNs) and precision (fewer FPs).
- VGG19: The single-stream VGG19 model (Fig. 8c) records 329 TP and 99 TN, with 57 FN and 40 FP. In the dual-stream version (Fig. 8d), TPs rise to 339 and TNs to 109, while FNs and FPs drop to 47 and 30, respectively. These changes show an overall improvement in the detection accuracy and reduction in misclassification rates.
- AlexNet: The single-stream AlexNet model (Fig. 8e) performs with 343 TP and 81 TN, alongside higher FNs (75) and FPs (26). The dual-stream setup (Fig. 8f) achieves better results, with TPs increasing to 353 and TNs to 91, while FNs and FPs decrease to 65 and 16. This improvement highlights a boost in both the recall and precision metrics for the dual-stream version.

Moreover, computational performance was measured in GFLOPS (Giga Floating Point Operations Per Second). It measures how many billion floating-point operations the model can perform per second (see Eq. (11)), providing a key indicator of its ability to handle intensive computations.

$$\text{GFLOPS} = \frac{\text{Total Floating Point Operations}}{\text{Inference Time (in seconds)}} \times 10^{-9} \quad (11)$$

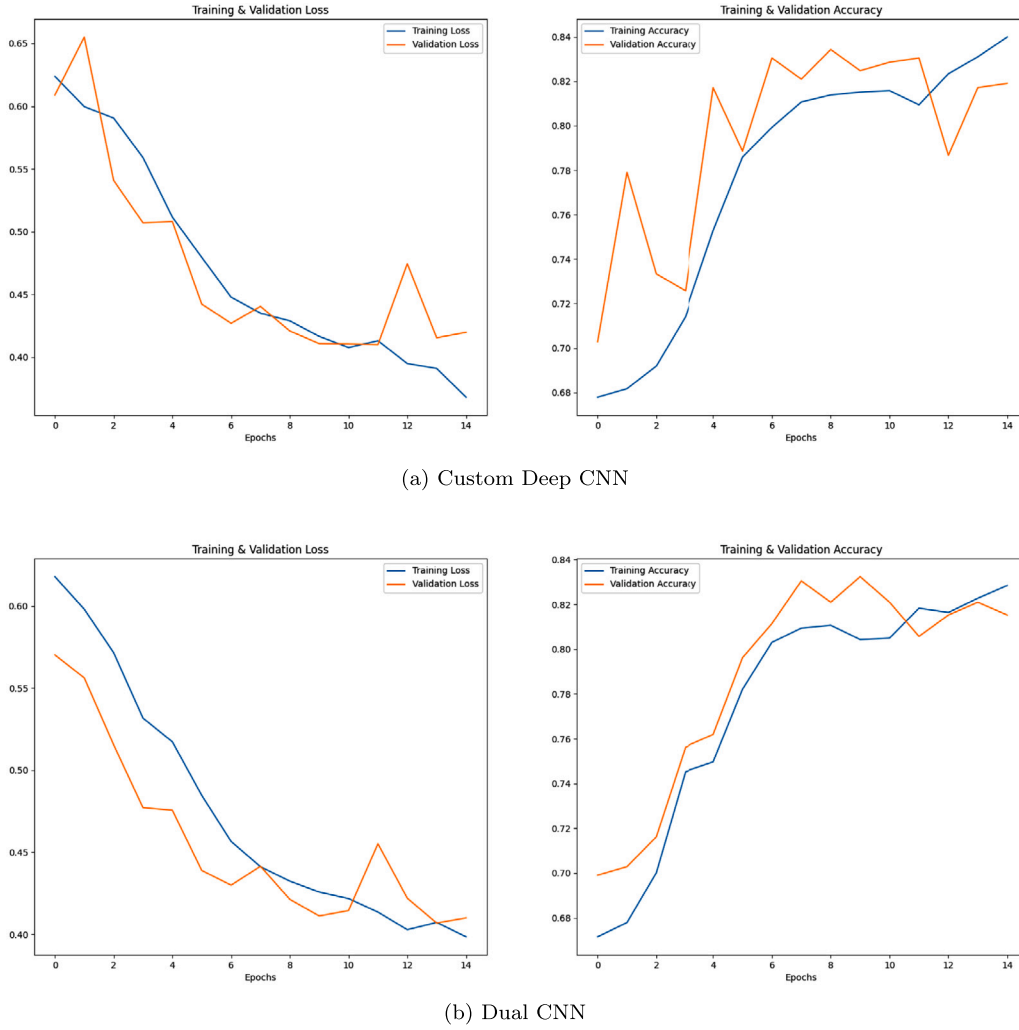


Fig. 5. Training and validation loss and accuracy over the epochs of: (a) the deep custom CNN stream, (b) our proposed Dual CNN model with both shallow stream and deep stream.

where *InferenceTime* is the total time a model takes to perform inference on a given input in seconds.

Table 6 presents a comparative analysis of single and dual stream CNN architectures in terms of parameters, GFLOPS, accuracy, recall, and F1 score for microcrack detection in PV cell images.

In the Single Stream Setup, three models were evaluated: Custom Deep CNN, AlexNet, and VGG19. The Custom Deep CNN, with 924,897 parameters and a GFLOPS of 0.07, achieved an accuracy of 80.95%, with a recall and F1 score both at 67.95%. AlexNet, with a significantly larger parameter count (87,622,337) and GFLOPS (0.55), reached a similar accuracy of 80.76%, but lower recall and F1 scores (51.92% and 61.6%, respectively). VGG19, at 21,234,689 parameters and 0.02 GFLOPS, showed the highest accuracy (81.52%) among the single-stream models, with recall and F1 scores of 63.46% and 67.12%.

In the Dual Stream Setup, the models were modified to enhance feature extraction by using parallel streams. The Dual Custom Deep CNN showed a significant improvement with an accuracy of 85.33%, recall of 71.71%, and F1 score of 73.9%, with an increase in parameters to 204,610,689 and a GFLOPS of 0.18. Similarly, the Dual AlexNet reached an accuracy of 84.57% and an improved F1 score of 69.2%, though recall remained moderate at 58.33%. Finally, the Dual VGG19, with 64,909,505 trainable parameters and 0.07 GFLOPS, matched the Dual Custom Deep CNN in accuracy (85.33%) and recall (69.87%), also achieving an F1 score of 73.9%.

This table highlights the dual stream setup's clear advantage in both accuracy and F1 scores, indicating improved detection capabilities,

especially for the Dual Custom Deep CNN and Dual VGG19, while illustrating the trade-offs in computational resources.

In addition, Fig. 9 shows the ROC (Receiver Operating Characteristic) curves. The ROC curves offer a comprehensive view of the dual CNN models' improved selectivity and robustness in distinguishing between true defects and false positives. For each model variant, whether based on a custom deep CNN, VGG19, ResNet50, or AlexNet architecture, the dual CNN approach consistently demonstrates higher true positive rates at lower false positive rates compared to their single-stream counterparts. In the ROC curves, this improvement is reflected by a larger area under the curve (AUC), signaling the model's strong discriminatory capability. The dual CNN structure leverages complementary feature information from its two streams: one capturing detailed, fine-grained textures and the other processing more abstract, high-level structural features. This synergy enhances the model's ability to filter out noise and avoid incorrectly labeling background variations or minor artifacts as defects.

As a result, the ROC curves show a reduction in false positives, indicating that the model is not only more sensitive to actual microcracks but also better at ignoring irrelevant or misleading patterns. This selectivity minimizes unnecessary alerts and improves the overall reliability of the detection system, making the dual CNN a more precise tool for quality inspection in PV applications.

Fig. 10 illustrates qualitative results of predictions made by the proposed dual CNN-based models. Figs. 10(a) and 10(b) depict examples where the model has successfully labeled the images, reflecting

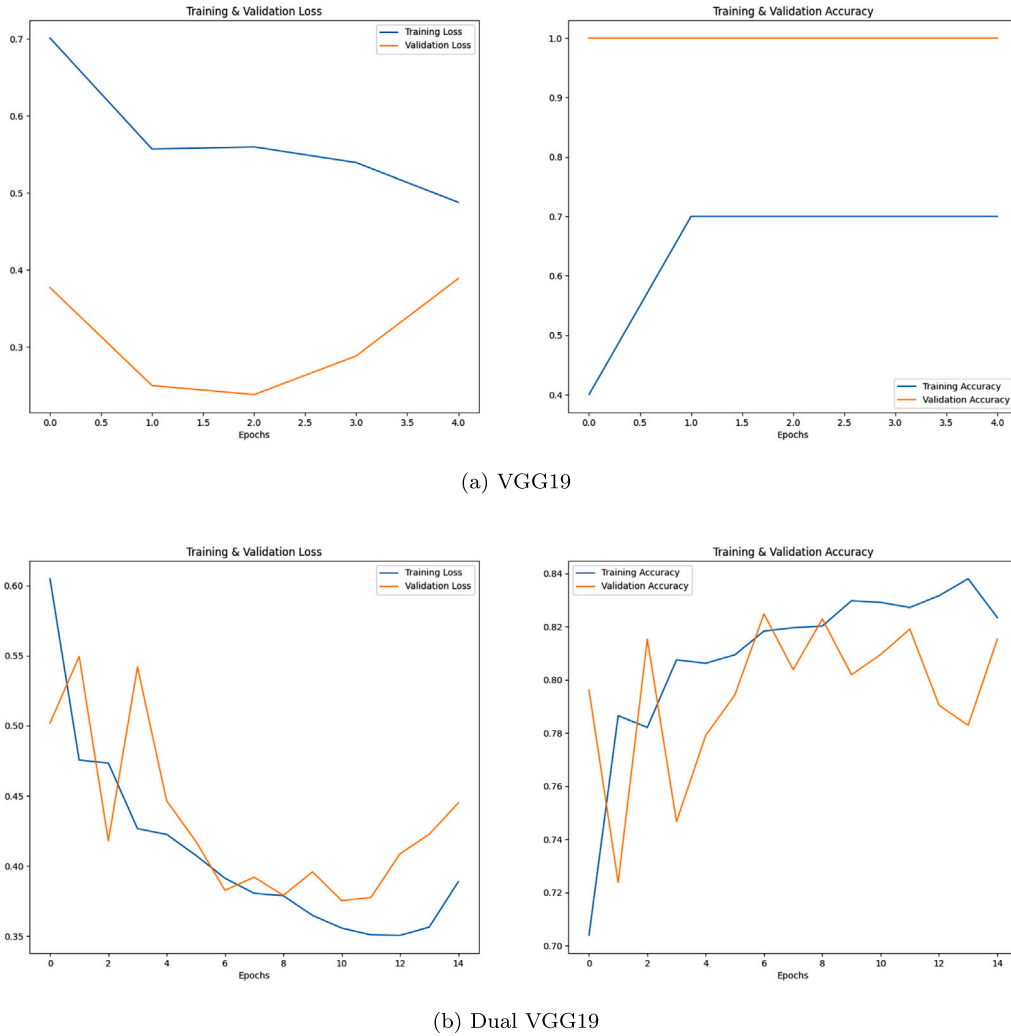


Fig. 6. Training and validation loss and accuracy over the epochs of: (a) VGG19 in single stream, (b) VGG19 in Dual stream setup.

Table 6

Performance comparison between single stream setup and dual stream setup for different CNNs.

Model	Nb of Param.	GFLOPS	Accuracy(%)	Recall	F1 Score (%)
Single Stream Setup					
Custom Deep CNN	924,897	0.07	80.95	67.95	67.95
AlexNet	87,622,337	0.55	80.76	51.92	61.60
VGG19	21,234,689	0.02	81.52	63.46	67.12
Dual Stream Setup					
Dual Custom Deep CNN	204,610,689	0.18	85.33	71.71	73.9
Dual AlexNet	131,296,577	0.64	84.57	58.33	69.2
Dual VGG19	64,909,505	0.07	85.33	69.87	73.9

its ability to capture relevant features for accurate classification. However, some images were misclassified, leading to false positives (FP), where defect-free cells are incorrectly identified as defective, and false negatives (FN), where defected cells are incorrectly identified as defect-free, as shown in Figs. 10(c), 10(d) for FP, and 10(e), 10(f) for FN. In particular, our model shows overly sensitization to certain image characteristics, potentially mistaking noise or minor inconsistencies for defects, more precisely:

- Fig. 10(c) shows an example of a PV EL image presenting surface irregularities and noise. These minor variations in brightness and sensor noise, introduced during image acquisition, are mistakenly interpreted by our model as defects.

- Fig. 10(d) shows an example of a PV EL image presenting textural patterns. Our model confused naturally occurring patterns, such as inherent material inconsistencies, with actual cracks.
- Fig. 10(d) shows an example of a PV EL image presenting edge artifacts where scratches, and electrode structures are mistaken for microcracks, leading to false alarms.

On the other hand, the missed defects are typically fine microcracks with low contrast or are located near cell borders where edge effects reduce visibility and hence our model struggled to detect. In particular:

- Fig. 10(e) shows an example of a PV EL image with low contrast where the crack blends into the background due to poor lighting

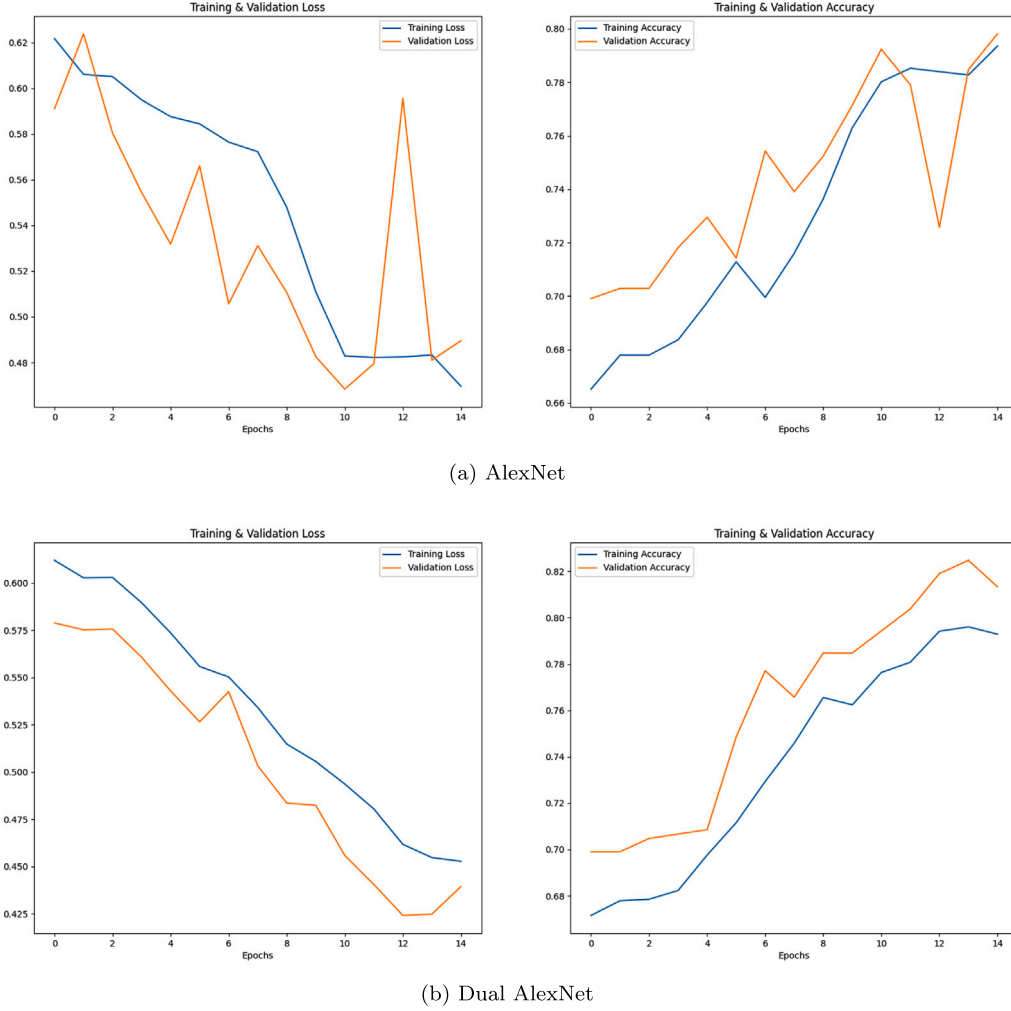


Fig. 7. Training and validation loss and accuracy over the epochs of: (a) AlexNet in single stream, (b) AlexNet in Dual stream setup.

conditions or inherent material properties. These properties did not stand out enough for our model to detect.

- Fig. 10(f) shows an example of a PV EL image with subtle micro-cracks where the defect is too fine that was hard for our model to detect.

To mitigate FPs, some advanced preprocessing techniques such as contrast normalization and denoising to suppress irrelevant variations could be introduced. Another strategy consists in incorporating post-classification verification using rule-based filters or lightweight anomaly detection modules to reduce FP rates.

To mitigate FNs, it would be beneficial to improve annotation quality through expert labeling and consensus-based review to ensure subtle defects are correctly captured during training.

5.5. Interpretability results

To enhance interpretability, we applied Grad-CAM [38] to visualize the model's attention on selected input samples. Fig. 11 presents the resulting heatmaps overlaid on examples of input images. The highlighted regions indicate spatial areas that contribute most to the model's decision, providing insight into how the dual-stream architecture highlights complementary features. These visualizations offer a preliminary yet valuable explanation of the learned representations, aligning with our planned future work on explainable AI.

5.6. Comparison with existing methods

Table 7 presents a performance analysis between our proposed dual-stream CNN models and existing state-of-the-art methods for defect detection in PV images. The table is segmented by dataset types, including infrared aerial images and electroluminescence (EL) images, highlighting each model's accuracy, F1 score, and computational complexity.

Among existing models, Shaik et al. [21] and Mamdouh et al. [24] achieved the highest accuracies with infrared aerial images, reaching around 98% but with high complexity, suggesting significant computational demands. On EL datasets, Hussain et al. led with a notable 97.4% accuracy and F1 score of 97, however is has high complexity.

Our dual-stream models, particularly the dual custom CNN, reached a respectable accuracy of 85.33% and 84.2% on EL images from [36] and from [37], respectively, with a lower complexity, offering a more efficient option for similar applications. While our approach shows a slightly lower accuracy and F1 score compared to some high-complexity models, it provides a balance of computational efficiency and accuracy. This trade-off makes our model a viable candidate for efficient PV defect detection, especially in resource-constrained environments where high-complexity models may not be feasible.

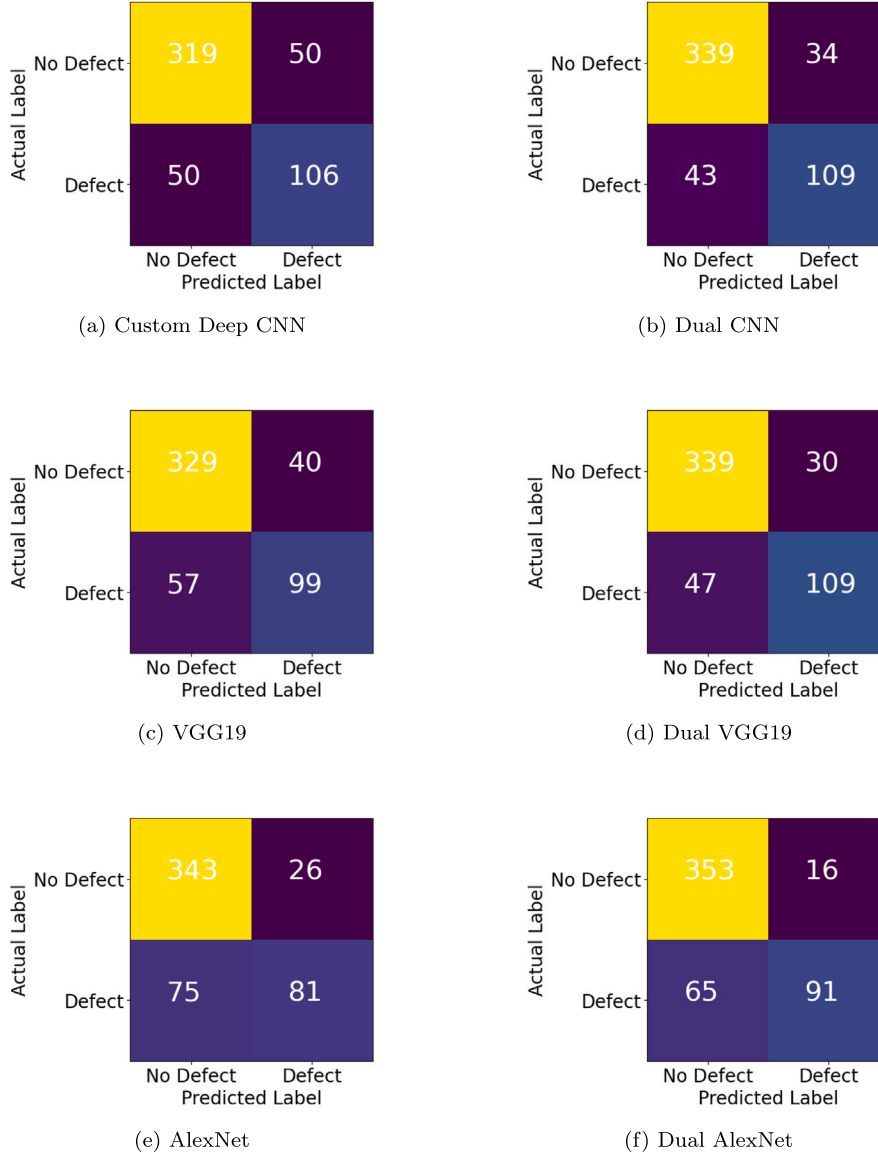


Fig. 8. Confusion matrices showing the number of True Positives (TP), and True Negatives (TN) on the first diagonal, and the False Negatives (FN) and False Positives (FP) on the second diagonal for the (a) single-stream custom deep CNN, (b) our proposed dual CNN classification model, (c) VGG19 in single stream, (d) VGG19 in Dual stream setup, (e) AlexNet in single stream, (f) AlexNet in Dual stream setup.

6. Discussion

The proposed dual CNN classification model shows improvements in identifying true positives and reducing false positives compared to the single-stream deep CNN model. However, this model also results in a higher number of false negatives and a decrease in true negatives. Particularly, the dual CNN model achieves 18 more true positives than the single-stream model (339 vs. 321). This indicates an improvement in the model's ability to correctly identify positive cases, suggesting that the dual CNN architecture is more effective at detecting the target defects. Our model has also fewer false positives (30 vs. 48), showing that it makes fewer incorrect positive predictions. This suggests that the dual CNN model might be better at avoiding false alarms, thus potentially improving its precision. The increase in true positives and the decrease in false positives highlight the dual CNN model's enhanced ability to detect defects.

The dual-stream architecture highlights on the strengths of both shallow and deep layers, with the shallow stream capturing fine-grained, texture-based details essential for identifying subtle, localized

defects. In contrast, the deep stream focuses on extracting coarse, high-level structural information across broader regions of the image. This combination ensures that the model is not only sensitive to minute variations within the PV cells—crucial for microcrack detection—but also capable of grasping contextual patterns that single-stream models often miss.

Moreover, we train the model using various noise augmentation techniques, such as rotation, Gaussian noise, salt-and-pepper noise, and contrast variations, ensuring robustness to real-world imaging conditions. In addition, By combining local and global features, the model can effectively suppress irrelevant noise while retaining crucial defect-related features, improving detection accuracy.

Ultimately, this enables the dual CNN to achieve more accurate, robust, and efficient detection of microcracks and defects, presenting a powerful tool for enhancing quality control in photovoltaic systems.

Moreover, to support the significance of performance differences between single-stream and dual-stream CNN models, we use a paired t-test. We conduct the paired t-test on the accuracy metric obtained from five independent runs of the dual CNN and single-stream CNN models. The test revealed a statistically significant improvement ($p < 0.05$),

Table 7

Comparison table of the performance metrics between the existing related works and our proposed method. Some metrics were not provided by the authors in their publications.

Method	Dataset	Accuracy (%)	F1 score(%)	Complexity
Duranay [20]	Infrared aerial Images	93.93	89.82	High
Shaik et al. [21]		98	99	High
Mamdouh et al. [24]		98.5	–	High
Wang et al. [25]		94.8	–	High
Nguyen et al. [26]	EL Images	83.11	84.3	Medium
Deitsch et al. [28]		88.42	84	Medium
Hussain et al. [29]		97.42	97	High
Rahman et al. [30]		96.97	–	Medium
ViT		82.27	80.04	High
CrossViT		84.32	81.14	High
ConvNext		81.12	80.01	High
Our networks				
Dual custom CNN	EL Images	85.33	73.9	Low
Dual VGG19	from [36]	85.33	73.9	Medium
Dual AlexNet		84.57	69.2	Medium
Dual custom CNN	EL Images	84.2	72.7	Low
Dual VGG19	from [37]	84.1	72.6	Medium
Dual AlexNet		83.44	68.8	Medium

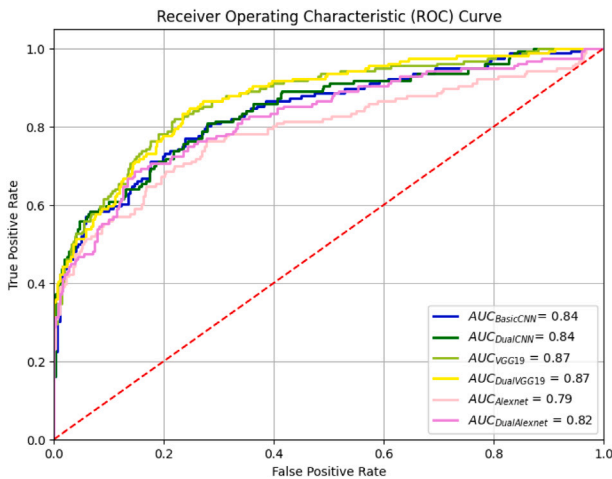


Fig. 9. Receiver Operating Characteristic (ROC) curves of the single-stream Deep CNN model, Dual CNN, VGG19, Dual VGG19, AlexNet, Dual AlexNet.

confirming that the observed performance gains are not due to random variation.

In addition, the performance of the dual models was evaluated using GFLOPS to assess computational efficiency alongside classification accuracy. The dual models achieved a computational throughput of an average of 0.6 GFLOPS. This demonstrates the balance between model complexity and computational demands, making it suitable for real-time defect detection in photovoltaic images. The combination of high classification accuracy and moderate GFLOPS showcases the model's practicality for deployment in environments with limited computational resources.

The proposed dual CNN architecture is well-suited for integration into real-world PV inspection systems. Its moderate computational demand makes it feasible for deployment on low-power edge devices, which are commonly used in industrial settings where real-time defect detection is critical. Despite introducing additional computational steps compared to traditional single-stream CNNs, the dual-stream design is optimized to maintain inference within practical bounds, ensuring minimal latency and enabling the simultaneous analysis of multiple PV cells. The architecture strikes a balance between accuracy and efficiency, making it scalable for automated inspection pipelines that process large image datasets on-site or in cloud-assisted environments.

In a complete inspection system, the dual CNN would operate as a core component in a multi-stage pipeline that includes image acquisition (e.g., electroluminescence or infrared imaging), preprocessing (noise reduction, alignment), and defect localization. The dual streams — shallow for fine-grained feature detection and deep for high-level semantic understanding — offer complementary advantages that enhance detection robustness across varied defect types. Practical challenges such as image quality variability, non-uniform lighting, and real-time throughput requirements will necessitate further field validation and potentially domain adaptation techniques. Nonetheless, the model's design is inherently compatible with scalable, high-throughput inspection systems needed for modern PV manufacturing and maintenance workflows.

7. Conclusion

In conclusion, the dual-stream approach enhances PV defect detection by effectively combining fine-grained details from shallow features with broader, context-rich deep semantic information. This integration leads to increased true positive and true negative counts across tested CNN models, demonstrating improvements in both sensitivity and precision over single-stream counterparts. It also shows a significant reduction in false positives and false negatives, underscoring its ability to detect subtle defects, like microcracks, while minimizing misclassifications. While the proposed Dual-CNN approach enhances microcrack detection in EL PV image, the reliance on labeled datasets for supervised training may introduce biases and limit generalization to unseen defect patterns or new PV technologies. In particular, real-world deployment of the dual CNN model for PV defect detection faces several challenges, including variability in imaging conditions, limited computational resources on edge devices, and the need for high-quality annotated data. Additionally, integrating the model into existing inspection systems and maintaining performance under high-throughput manufacturing conditions can be complex. Addressing false positives and negatives in diverse real-world scenarios remains crucial to ensure reliability and minimize costly misclassifications. For industrial implementation, the model must meet key requirements such as real-time inference speed. It should also support scalability for high-volume inspection, comply with industry standards for reliability and accuracy, and operate efficiently on hardware-constrained environments typical in manufacturing settings.

As a future work, we will interpret the obtained results using XAI tools such as Grad-CAM. Moreover, we intend to enhance the energy efficiency of the proposed model using quantum-inspired computing algorithms. Moreover, the study can be extended to include additional

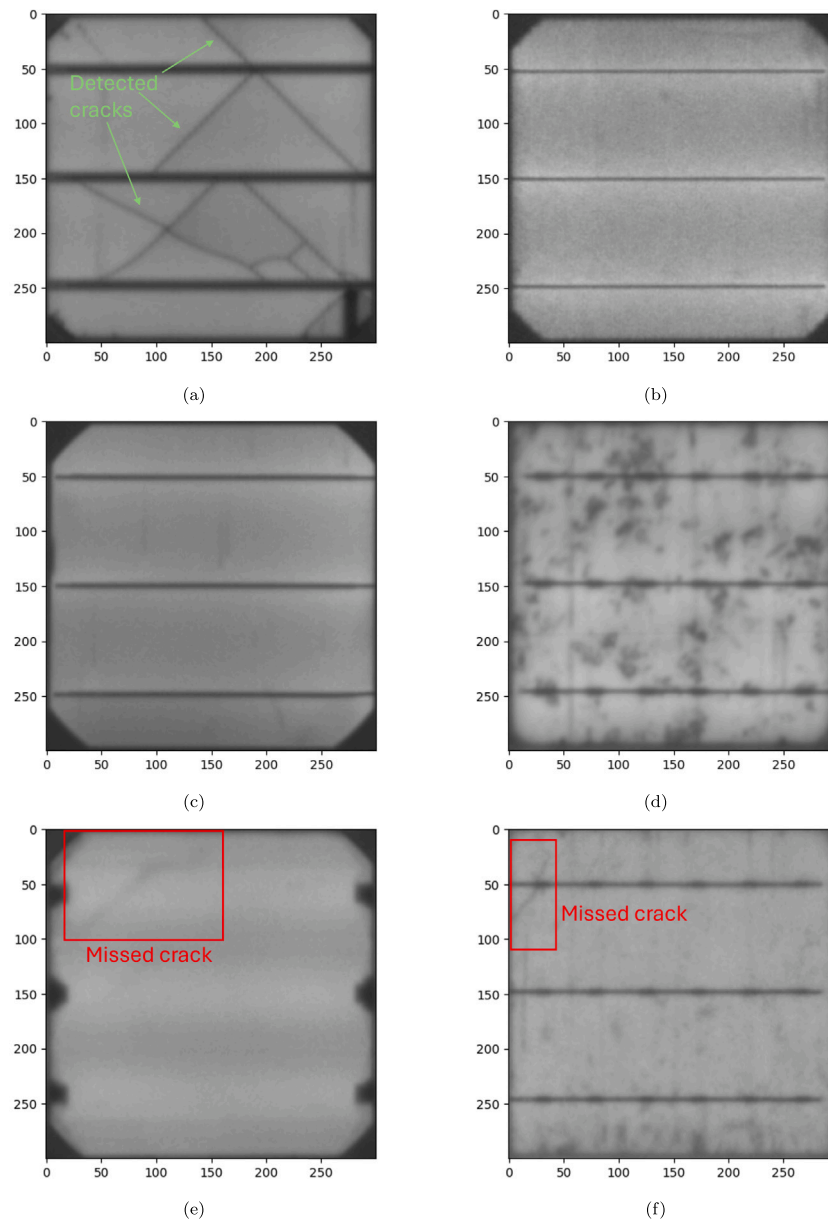


Fig. 10. Qualitative results of predictions made by the proposed CNN with correctly classified solar cell images: (a) TP, (b) TN and misclassifications: (c), (d) FP and (e), (f) FN.

defect types beyond microcracks, such as discoloration, broken fingers, snail trails, and PID (Potential Induced Degradation). Finally, to improve the model's efficiency and suitability for deployment, various model optimization techniques can be investigated. These include pruning, which reduces model size by removing less significant weights; quantization, which lowers the precision of weights and activations to reduce memory and computational load.

CRediT authorship contribution statement

Khoulood Samrouth: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Souha Nazir:** Writing – review & editing, Writing – original draft, Validation. **Nader Bakir:** Resources, Data curation. **Nadine Khodor:** Writing – review & editing, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

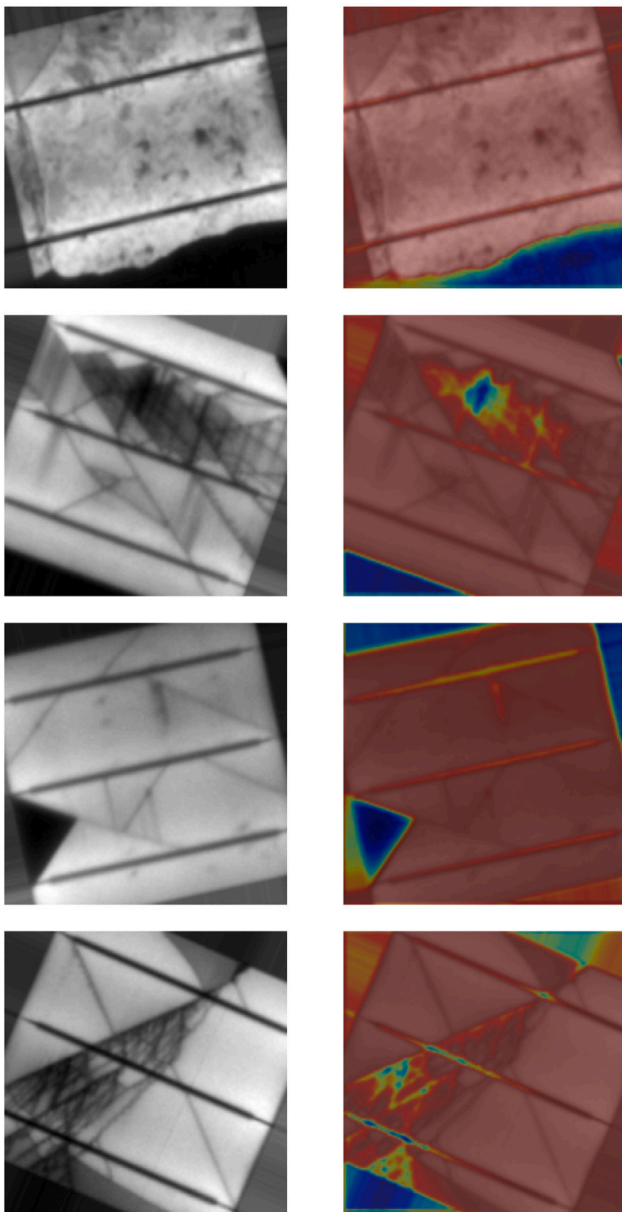


Fig. 11. Grad-CAM heatmap visualizing feature attention of the dual-stream CNN. The model correctly classifies the samples (left), focusing on discriminative regions of the input. The highlighted areas (right) reveal which spatial features influence the prediction most strongly.

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